

NQDP: A Query-Based Diagnostic Protocol for IPv4-64

Replacing ICMP’s diagnostic role with a fixed-size, trust-gated, carrier-controlled query service that reveals nothing to attackers and everything to the operators who are entitled to it

Registry: [?]

Series Path: [\[HOWL-NET-1-2026\]](#) → [\[HOWL-NET-2-2026\]](#) → [\[HOWL-NET-3-2026\]](#) → [\[HOWL-NET-4-2026\]](#) → [?]

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1. Purpose

This document specifies NQDP, the Network Query Diagnostic Protocol. NQDP is the diagnostic service for IPv4-64, the protocol defined in the companion documents [\[HOWL-NET-1-2026\]](#) through [\[HOWL-NET-4-2026\]](#).

IPv4-64 deliberately removed ICMP’s role from the forwarding plane. The forwarding plane answers no questions. It drops malformed, undeliverable, or unrecognized packets in silence. This is a security decision, explained in Section 2. But diagnostics are necessary. Operators must be able to ask whether a destination is reachable, what a route looks like, and which instance of an anycast service they are hitting. NQDP provides that capability as a separate service that runs above the forwarding plane, not inside it.

NQDP is a proposal. It is not a plan, not a mandate, and not a replacement for any deployed system. It exists because IPv6, for all that it advanced the art, is not desirable in many networks, and the diagnostic model built into the IP layer — ICMP — carries costs that a modern design should not accept. NQDP keeps the diagnostic capability operators need while removing the attack surface that capability has always carried.

2. Background

2.1 What ICMP Is

ICMP is the Internet Control Message Protocol. It is the part of IP that generates diagnostic and error messages. When a router drops a packet because its time-to-live reached zero, it sends an ICMP “Time Exceeded” message back to the sender. When a host is asked for a service it does not run, it sends an ICMP “Port Unreachable.” The `ping` command works by sending an ICMP “Echo Request” and waiting for an “Echo Reply.” The `traceroute` command works by sending packets with increasing time-to-live values and reading the ICMP “Time Exceeded” messages that come back from each hop.

ICMP was designed in 1981, when every host on the network was trusted. It was built into the IP layer so that any router or host could answer a diagnostic query as a side effect of normal forwarding. This made debugging easy.

2.2 Why ICMP Is a Problem

The same property that makes ICMP useful to an operator makes it useful to an attacker. Every ICMP response confirms that a host exists. A scanner sends a probe to an address; if an ICMP reply comes back, the scanner knows a live host is there. The content of the reply often reveals which operating system the host runs, because different systems generate different time-to-live values and message formats. This is reconnaissance, and ICMP hands it out for free to anyone who asks.

ICMP also shares a channel with production traffic. It operates at the IP layer, the same layer that forwards real data. There is no clean way to say “answer diagnostic queries from my operations team but not from the internet,” because the forwarding plane itself emits the responses. An operator can configure a firewall to suppress some ICMP, but that is a patch applied per host or per network, not a property of the protocol.

The companion documents established the principle that a modern forwarding plane should reveal nothing and answer nothing. A malformed packet is dropped in silence. A scan receives no response. This closes the reconnaissance surface completely. But it also removes `ping`, `traceroute`, and path-MTU discovery, because those tools depend on the forwarding plane answering. NQDP restores the diagnostic capability without restoring the surface.

2.3 The Design Inversion

ICMP answers anyone, reveals host existence, shares a channel with production traffic, and generates responses as a side effect of forwarding. NQDP is the opposite on every axis.

NQDP answers only the parties the carrier chooses. It reveals only the facts the carrier chooses to disclose. It runs as its own service, not as a side effect of forwarding. And it generates a response only on an explicit, authenticated request. The capability is the same — tell me about reachability and paths — but the control over who learns what belongs entirely to the carrier.

3. What NQDP Is

NQDP is an application-layer protocol. It runs as ordinary payload over IPv4-64 UDP, on port 56789. It is not part of the forwarding plane. The forwarding plane carries NQDP packets exactly as it carries any other UDP traffic, through the same seven phases, under the same security gates, with no special handling.

A requester sends one fixed-size request naming a target prefix. A responder sends back one fixed-size response containing every diagnostic fact it is willing to disclose about that target, computed at one instant. One request, one response, both of fixed size, both known in full before a byte moves.

The facts NQDP carries are the ones network operators actually ask for: the cost to reach a target, the origin autonomous system seen for it, the next-hop autonomous system, a summary of the path, whether the route is RPKI-valid, whether the target is reachable, how long its route has been stable, and which instance of an anycast service was reached. Each fact is a single number. None of them is the full network path, because the full path is topology, and disclosing topology is the reconnaissance problem NQDP exists to avoid.

3.1 Terms Used in This Document

Autonomous System (AS). A network under one administrative control, identified by a number. BGP routes traffic between autonomous systems.

BGP. The Border Gateway Protocol, which routers use to exchange reachability information across the internet. A router's BGP table holds the routes it knows and the cost it assigns to each.

AS_PATH. The list of autonomous systems a route passes through. It is the standard way to understand where traffic goes and how it gets there.

RPKI. Resource Public Key Infrastructure, a system for cryptographically validating that an autonomous system is authorized to originate a route. An RPKI-valid route is trustworthy; an RPKI-invalid route may be a hijack.

Anycast. A service deployed at many locations that share one address. Traffic to that address is delivered to whichever instance is nearest. Anycast makes it hard to tell which instance answered.

SVT. The Source Validation Token from [\[HOWL-NET-1-2026\]](#). It is a cryptographic stamp a router applies that lets a receiver verify the source of a packet. In NQDP, the SVT determines whether a requester is trusted.

Fixed offset. A field is at a fixed offset when it always sits at the same byte position in every packet. There is nothing to calculate and nothing to parse wrong.

Sentinel. A reserved value that means “no answer here.” In NQDP the sentinel is always all-ones for the field's width.

4. The Core Design Choices

NQDP makes five choices that shape everything else. Each one is explained here before the packet formats, so the formats read as consequences rather than decisions.

4.1 Everything Comes Back, Always

A requester cannot ask for a subset of facts. It sends one request and receives the complete record every time. This has three consequences.

First, there is exactly one request shape and one response shape, so there is exactly one implementation. A protocol that let you select fields would have many possible response shapes, and implementations would diverge on how they handle the ones you did not ask for.

Second, the response is a compile-time constant size, identical regardless of what was answered. An observer cannot tell a rich answer from an empty one by looking at the length.

Third, the response drops directly into a time series. Every field is always present at the same offset, so each response is one row with every column populated. Analyzing change over time becomes trivial: compare this row to the last.

4.2 Withheld and Unknown Look the Same

A field the carrier chooses not to disclose carries the sentinel. A field the responder genuinely cannot determine also carries the sentinel. On the wire they are identical. This is deliberate. If the packet announced “I am withholding this,” an attacker could poll a responder and map exactly what each carrier will and will not disclose. By making withheld and unknown indistinguishable, NQDP keeps the carrier’s disclosure policy unprobeable. The requester, who knows its own relationship with the carrier, can interpret a sentinel in context; a stranger cannot.

4.3 The Responder Reads Nothing the Requester Claims About Itself

The only requester-supplied values the responder acts on are the target prefix and its address family. The requester sends no timestamp, no sequence number, and no claim about its own identity. Anything an attacker could poison by supplying false data simply does not exist on the wire. The responder supplies its own time, from its own clock, which never moves backward. Trust is not asserted by the requester either; it is established by the SVT layer below NQDP and inherited as a single bit.

4.4 The Hard Work Happens in the Background

The responder never computes a diagnostic value when a request arrives. Every value — the cost, the path summary, the stability time — is computed in advance by the control

plane, which watches the carrier's BGP table and updates the values whenever a route changes. These pre-computed values are injected into a table that the fast data plane reads. When a request arrives, the data plane does one table lookup and assembles the response. It reads a cached answer; it does not produce one.

This is what lets a diagnostic service run at line rate. The expensive reasoning about routes happens at BGP's pace, which is seconds to minutes, decoupled from any request. The request triggers a lookup, never a computation. An attacker cannot craft an expensive query because no query is expensive — every one costs the same single table read.

4.5 The Cost Is the Carrier's Own BGP Cost

NQDP does not define a cost model. The cost a responder returns is taken directly from the carrier's own BGP table. The carrier sets this cost by setting its BGP policy, exactly as it already does. A requester reading the cost is reading the carrier's real routing preference. Lower means the carrier prefers that path. The requester compares costs from different responders and routes toward the lower one. The carrier drives the requester's traffic by publishing a number it fully controls, in its own units, which reveal nothing about how the number was derived.

5. The Request Packet

The request is 24 bytes, fixed, six 32-bit words. It names a target and carries an identifier the responder will echo.

offset	width	field	purpose
0	64	request_id	client-chosen identifier, echoed verbatim.
8	64	target_prefix	the address or prefix being asked about.
16	8	target_prefix_len	significant bits of target_prefix.
17	8	addr_family	0 native, 1 IPv4-compat, 2 IPv6 /64 prefix.
18	16	reserved	zero in v1.
20	32	reserved	zero in v1.

total: 24 bytes

The `request_id` is how a requester matches a response to a request. The responder copies it back unchanged. The requester records when it sent the request and when the matching response arrived, both on its own clock, and the difference is the round-trip time. No time travels on the wire from the requester, because the responder has no reason to read it and every reason not to.

The `addr_family` tells the responder how to read `target_prefix`: as a native IPv4-64 address, as a legacy IPv4 address in the low 32 bits, or as an IPv6 /64 prefix for transit traffic. This matches the address vocabulary of [HOWL-NET-4-2026].

There is no credential field. Whether the requester is trusted is decided by the SVT layer below NQDP, before NQDP logic runs, and is inherited as one bit.

6. The Response Packet

The response is 48 bytes, fixed, twelve 32-bit words. Every field is always present. A field that is withheld or unknown carries the all-ones sentinel.

offset	width	field	sentinel	meaning
0	64	<code>request_id</code>	—	echoed from the request.
8	32	<code>responder_epoch</code>	—	responder clock at computation.
12	32	<code>reserved</code>	—	zero in v1.
16	32	<code>cost</code>	0xFFFFFFFF	BGP cost to reach target. Lower prefer
20	32	<code>origin_as_seen</code>	0xFFFFFFFF	origin AS seen. Compare for hijack
24	32	<code>next_hop_as</code>	0xFFFFFFFF	immediate next-hop AS.
28	32	<code>as_path_hash</code>	0xFFFFFFFF	plain hash of the AS_PATH.
32	8	<code>as_path_len</code>	0xFF	number of AS hops.
33	8	<code>rpki_status</code>	0xFF	0 valid, 1 invalid, 2 not-found.

34	8	reachable	0xFF	0 no, 1 yes, 2 routable, host unconfirm
35	8	scrubbing_active	0xFF	0 no, 1 yes. Traffic via scrubbing p
36	32	stability_epoch	0xFFFFFFFF	time of last route change.
40	16	anycast_modulus	0xFFFF	which instance of the ring reached.
42	16	anycast_count	0xFFFF	ring size N.
44	32	reserved	—	zero in v1.

total: 48 bytes

6.1 The Binding Fields

`request_id` is the echo that ties this response to the question that was asked. `responder_epoch` is the responder’s clock at the instant it computed the record. It is the timestamp for the requester’s time series. It is never checked against anything the requester sent, because the requester sends no time. The responder’s clock is monotonic and never goes backward, which is why NQDP needs no sequence number: time alone orders the records.

6.2 The Routing Fields

`cost` is the carrier’s BGP cost, explained in Section 4.5. `origin_as_seen` is the autonomous system the responder sees as the origin of the target’s route; comparing it to the expected origin detects a hijack or route leak. `next_hop_as` is the immediate next autonomous system the responder would hand the traffic to — one hop of the path, not the path.

`as_path_hash` is a plain hash of the full `AS_PATH`. It is plain, not keyed, so that two responders on the same path produce the same hash. This lets an operator compare answers from two vantage points: the same hash means the same path, a different hash means the paths diverge, which reveals asymmetric routing. A changed hash across two polls from one responder means the path changed. The hash discloses that two paths are the same or different; it never discloses what the path is. `as_path_len` is the number of hops, which allows route comparison without revealing topology.

`rpki_status` reports whether the route is RPKI-valid, invalid, or not found. `reachable` reports whether the target can be reached, with a distinct value for “routable, but host existence not confirmed” so that reachability can be answered without confirming a specific host exists. `scrubbing_active` reports whether traffic to the target currently flows through a DDoS-scrubbing path, which lets an operator confirm

that mitigation is in effect.

6.3 The Stability Field

`stability_epoch` is the responder's clock value at the last route change for the target. Subtracting it from `responder_epoch` gives how long the route has been stable. NQDP carries no flap count, because a count is a rate, and a rate cannot be known from a single snapshot. An operator polling over time watches `stability_epoch` and `as_path_hash` change and derives the flap frequency in their own time series. The protocol carries single-instant truth; rates are the requester's own discovery.

6.4 The Anycast Fields, Which Also Replace Ping

`anycast_modulus` is which instance of an anycast ring the requester reached. `anycast_count` is the size of the ring. Together they say "you reached instance 7 of 8."

This is also the modern replacement for ping. Reaching an instance is the confirmation that the service is alive. A target that is a single host is a ring of one: a response of "1 of 1" means you reached it, which is exactly what a successful ping confirmed. A response of "7 of 8" means you reached a live member of a larger service. The liveness confirmation is the fact that an instance answered, without the host-existence reconnaissance that ICMP Echo leaked, because you learn you reached a member of a known service, not that a specific hidden host exists.

The full ring — every instance and its cost — is never returned in one packet, because a ring of N instances is a variable-length structure and NQDP carries nothing variable. Each response describes only the one instance reached. An operator assembles the full ring picture in their time series, one row per instance as repeated polls land on different members. The time series is the ring table; the packet stays fixed.

6.5 Why Numbers Instead of Paths

Every field is a single number. This is the central discipline. A full `AS_PATH` is topology, and topology in the hands of an attacker is a map of the network. NQDP returns the diagnostic consequences of the path — its length, a hash that detects change and asymmetry, the origin and next-hop autonomous systems — without returning the path. An operator gets the answers they need for hijack detection, asymmetry detection, flap detection, and route comparison, each from a number, and the topology never leaves the responder.

7. Trust and Disclosure

7.1 Trust Is Inherited, Not Carried

NQDP does not authenticate anyone. The forwarding plane's SVT layer already determined whether the source of the packet is validated, and NQDP reads that result as a

single bit. A requester presenting a valid SVT is trusted; one that is not is public. There is no credential in the NQDP packet and no authentication step in the NQDP logic, because both already happened below.

7.2 Two Audiences, Two Configurations

The carrier defines two disclosure configurations: one for SVT-trusted requesters and one for the public. Each configuration is a bitfield with one bit per response field. A set bit means “replace this field with the sentinel for this audience.” The carrier can withhold any field from either audience independently. A carrier might disclose cost to the public but reveal the origin AS only to trusted peers; that is one bit in each configuration.

Both configurations are live data that the control plane can edit without recompiling anything. When a request arrives, the data plane selects the configuration by the trust bit, applies the sentinels it specifies, and emits the response. The disclosure decision is a single bitfield lookup.

7.3 Slam-Shut

A configuration can also be set to drop its entire audience. Under attack, a carrier sets the public configuration to drop. Public requesters then receive nothing, while SVT-trusted requesters keep receiving full answers. This is the moment pathing data is most valuable to peers, so the carrier sheds only the untrusted public and keeps answering the operators it trusts. Slam-shut is not a separate mechanism; it is the drop setting on the public configuration. There is no new state and no new code path.

7.4 Why There Is No Rate Limiting

A query service normally needs rate limiting because answering is expensive, and an attacker exploits that by making the responder do costly work cheaply. NQDP answering is a single table read, no more expensive than forwarding a packet. A responder built to forward at line rate can answer NQDP queries at line rate. There is no asymmetry to exploit, so there is nothing to rate-limit. The carrier expresses its intent as policy — answer fully, answer with sentinels, or drop — and serves that policy at line rate. A flood of queries is absorbed the same way a flood of data packets is absorbed, by the forwarding plane’s own protections, which gate volumetric attacks before they reach the NQDP port at all.

8. The Responder Lifecycle

The responder runs only for packets on UDP 56789. Every step is fixed and completes in constant time. The data plane reads injected tables; it computes nothing expensive.

```
D1 Parse      Extract the fixed 24-byte request at fixed offsets.
D2 Validate    reserved == 0, size exact. Else silent drop.
```

```

D3  Box mode      respond, relay, or drop (Section 9).

D4  Audience      read the inherited SVT bit.

D5  Lookup        one prefix lookup into the injected bundle. Miss → all sentinels

D6  Policy        select svt or public config; apply its sentinel bitfield;

                        if the config is set to drop this audience, stop.

D7  Assemble      copy the bundle, apply sentinels, echo request_id,

                        stamp responder_epoch from the local monotonic clock.

D8  Emit          serialize the fixed 48-byte response.

```

The data plane holds only the injected tables, the two disclosure bitfields, and its local clock. There is no connection state, no request history, no rate meter, no sequence counter. The expensive values come from the control plane, which watches the carrier's BGP table and, whenever a route changes, recomputes the affected prefix's bundle — cost taken verbatim from the BGP table, origin and next-hop autonomous systems, the path hash and length, the RPKI status, reachability, scrubbing state, and the time of the change — and injects the updated bundle into the table the data plane reads.

This is the same division of labor the forwarding plane already uses: the slow, general-purpose control plane computes and installs; the fast data plane matches and serves; the table is the contract between them. NQDP applies that established pattern to a different question.

9. Deployment

Every box can run NQDP. Each box is configured, per port, in one of three modes.

Respond. The box answers from its own injected table.

Relay. The box does not answer but forwards the query toward the next hop on the way to the target. The query follows normal routing until it reaches a box in respond mode, or, if the carrier chooses to be transparent, all the way to the destination.

Drop. The box refuses the query, and the requester receives nothing.

This is a basic configuration option present on all devices. A carrier deploys NQDP everywhere and decides, box by box, which ones answer, which ones pass queries along, and which ones refuse. The diagnostic table competes for memory with the forwarding tables on any box set to respond, so a carrier with tight table budgets sets most boxes to relay and concentrates responders where there is room.

Because relay forwards toward the target, a query naturally finds the box responsible for answering about that target without any extra configuration. The requester does not need to know in advance which box will answer; it addresses the target, and the network delivers the query to a responder on the path.

10. What NQDP Provides

NQDP restores the diagnostic capability that IPv4-64 removed from the forwarding plane, without restoring the attack surface that capability carried in ICMP.

It cannot be used for reconnaissance, because it reveals only the scalar facts a carrier chooses to disclose, and reaching an anycast instance confirms a service is alive without confirming a hidden host exists.

It cannot be amplified, because the request is 24 bytes and the response is 48 bytes, close to one-to-one, with no multiplier.

It cannot be used to probe a carrier's disclosure policy, because withheld and unknown fields are identical on the wire.

It cannot be flooded for advantage, because answering is a single table read at line rate, and a flood is absorbed the same way the forwarding plane absorbs any volumetric traffic.

It cannot be poisoned by attacker-supplied data, because the responder reads nothing the requester claims about itself — no time, no sequence, no identity — and supplies its own monotonic time instead.

And it gives operators the facts they actually need: reachability, cost, origin and next-hop autonomous systems, path length, path-change and asymmetry detection, RPKI validity, route stability, scrubbing confirmation, and anycast instance identity — each as a single number, each dropping directly into a time series, each disclosed only to the parties the carrier has chosen.

The diagnostic function that ICMP built into the forwarding plane, where it served friendlies and attackers alike, becomes a separate service that the carrier tunes to disclose what it wants, to whom it wants, at line rate, revealing nothing to anyone else.

11. What This Document Changes

Nothing in the companion documents changes. The forwarding plane, the IP header, the TCP and UDP headers, the security model, and the P4 implementation are unchanged. NQDP is a new application-layer service that rides over IPv4-64 UDP and inherits the SVT trust bit. It adds a diagnostic capability; it modifies no existing field, header, or forwarding behavior.

Appendix A — Request Byte Map

byte	field	
0-7	request_id	(64 bits)
8-15	target_prefix	(64 bits)
16	target_prefix_len	(8 bits)
17	addr_family	(8 bits)
18-19	reserved	(16 bits, zero in v1)
20-23	reserved	(32 bits, zero in v1)
Total: 24 bytes, 6 words.		

Appendix B — Response Byte Map

byte	field	
0-7	request_id	(64 bits)
8-11	responder_epoch	(32 bits)
12-15	reserved	(32 bits, zero in v1)
16-19	cost	(32 bits)
20-23	origin_as_seen	(32 bits)
24-27	next_hop_as	(32 bits)
28-31	as_path_hash	(32 bits)
32	as_path_len	(8 bits)
33	rpki_status	(8 bits)
34	reachable	(8 bits)
35	scrubbing_active	(8 bits)
36-39	stability_epoch	(32 bits)

40-41	<code>anycast_modulus</code>	(16 bits)
42-43	<code>anycast_count</code>	(16 bits)
44-47	<code>reserved</code>	(32 bits, zero in v1)

Total: 48 bytes, 12 words.

Appendix C — Sentinel Values

The not-available value of any field is all-ones for its width. Binding fields (`request_id`, `responder_epoch`) and `reserved` fields have no sentinel because they are always present and meaningful.

field	sentinel
<code>cost</code>	<code>0xFFFFFFFF</code>
<code>origin_as_seen</code>	<code>0xFFFFFFFF</code>
<code>next_hop_as</code>	<code>0xFFFFFFFF</code>
<code>as_path_hash</code>	<code>0xFFFFFFFF</code>
<code>as_path_len</code>	<code>0xFF</code>
<code>rpki_status</code>	<code>0xFF</code>
<code>reachable</code>	<code>0xFF</code>
<code>scrubbing_active</code>	<code>0xFF</code>
<code>stability_epoch</code>	<code>0xFFFFFFFF</code>
<code>anycast_modulus</code>	<code>0xFFFF</code>
<code>anycast_count</code>	<code>0xFFFF</code>

Appendix D — Field Origin

field	answers the operator question
<code>cost</code>	which path does the carrier prefer, and how much

<code>origin_as_seen</code>	is this prefix being originated by the right AS (hijack)
<code>next_hop_as</code>	who does the carrier hand my traffic to next
<code>as_path_hash</code>	did the path change, and is routing asymmetric
<code>as_path_len</code>	how long is the path, for comparison
<code>rpki_status</code>	is the route cryptographically valid
<code>reachable</code>	can the target be reached at all
<code>scrubbing_active</code>	is DDoS mitigation currently in the path
<code>stability_epoch</code>	how long has the route been stable (flap)
<code>anycast_modulus</code>	which instance did I reach (and modern ping)
<code>anycast_count</code>	how many instances are in the ring

Appendix E — Config Vocabulary

Per box, per NQDP port:

`mode` $\in$ { `respond`, `relay`, `drop` }

Per responder, two live bitfields:

`svt_config` one bit per response field; set = sentinel for SVT requesters

`public_config` one bit per response field; set = sentinel for public requesters

each config also has a whole-audience drop flag

`slam-shut` under attack = set `public_config.drop`

HOWL-NET-5-2026 — Supporting Appendices

These appendices supply detail the main specification references but does not carry inline: the operator-problem mapping that drove the field set, the full data-plane and control-plane contract, the time-series behavior that fields depend on, the attack-surface accounting, and the connections to the four prior documents. Nothing here restates the packet layout or the lifecycle from the main document; it carries the material those depend on.

Appendix F — Operator Problem → Field Mapping

Each real diagnostic need, the tool operators use today, what that tool leaks, and the NQDP field that answers the same need without the leak.

Operator need	Today's tool	What today's tool leaks or costs	NQDP field	How NQDP avoids the cost
Is the target alive	ICMP Echo (ping)	Confirms host existence to anyone; enables OS fingerprinting	<code>anycast modulus / anycast count</code>	Confirms a service instance answered, not that a hidden host exists
What path does traffic take	traceroute	Reveals every hop; shares channel with production	<code>as path hash, as path len, next hop as</code>	Returns path consequences as numbers, never the hop list
Is this route a hijack	external looking glass, RPKI lookup	Looking glasses are scraped HTML, no clean API	<code>origin as seen, rpki status</code>	Machine-readable scalar, one fixed response
Is routing asymmetric	traceroute from both ends, manual compare	Requires access to both source nodes	<code>as path hash from two vantages</code>	Same hash = same path; compared in the requester's series
Is the route flapping	BGP monitoring (BMP), route collectors	Firehose to a trusted collector; heavy infrastructure	<code>stability epoch over polls</code>	Single-instant stability time; rate derived in the series
Which carrier path is better	BGP cost, hidden inside the AS	Not externally visible at all	<code>cost</code>	Carrier's own BGP cost, published as one opaque scalar
Who gets my traffic next	BGP table access	Requires the source's BGP table	<code>next hop as</code>	One AS number, no table access needed

Operator need	Today's tool	What today's tool leaks or costs	NQDP field	How NQDP avoids the cost
Is DDoS mitigation active	manual confirmation with provider	No in-band signal	scrubbing active	One bit the carrier chooses to expose
Which anycast instance answered	no standard tool	Anycast is opaque by design	anycast modulus / anycast count	Position-in-ring as a modulus, no instance location

The mapping shows the field set is not invented. It is the enumerated set of facts operators already extract, by harder means, from looking glasses, BMP collectors, traceroute, and direct BGP access. NQDP collapses that scattered, leak-prone, access-dependent set into one fixed query answered by numbers.

Appendix G — NQDP Lifecycle vs Forwarding-Plane Lifecycle

NQDP reuses the seven-phase shape of the forwarding plane deliberately, so an implementer already fluent in [\[HOWL-NET-3-2026\]](#) reads NQDP as the same pattern pointed at a different question.

Forwarding phase (NET-3)	NQDP phase (NET-5)	Shared property
Parse (fixed offsets)	D1 Parse (fixed 24 bytes)	No variable-length parsing
Validate (strict drop)	D2 Validate (reserved, size)	Any failure → silent drop
Authenticate (SVT/cookie/token)	D4 Audience (inherit SVT bit)	SVT is the trust root; NQDP reuses it, does not recompute
Classify (connection/threat/firewall)	D6 Policy (disclosure bitfield)	A table/config decides treatment at a fixed key
Route (LPM on dst addr)	D5 Lookup (LPM on target prefix)	One longest-prefix match on a 64-bit key
Transform (stamp, meter, track)	D7 Assemble (sentinel, echo, stamp)	Fixed-field write, O(1)
Emit (fixed length)	D8 Emit (fixed 48 bytes)	Fixed-length serialize, no padding math

The one phase NQDP adds and the forwarding plane lacks is D3 Box mode (respond/relay/drop), because a forwarding node always forwards, whereas an NQDP node

chooses whether to answer. The one phase the forwarding plane has that NQDP lacks a per-request version of is Authenticate: NQDP does no cryptographic work per request because the SVT check already happened in the forwarding plane that carried the packet to the NQDP port.

Appendix H — Data-Plane / Control-Plane Contract

The table is the contract. The control plane writes; the data plane reads. This appendix lists every shared structure, who writes it, who reads it, and at what rate.

Structure	Written by	Written at rate	Read by	Read at rate
nqdp prefix (the bundle table)	nqdp bgp sync	per BGP route change (sec-min)	D5 Lookup	per request (line rate)
anycast modulus (this box's position)	nqdp anycast	once per served prefix	D5 Lookup	per request
anycast count (ring size)	nqdp anycast	per ring membership change	D5 Lookup	per request
svt config bitfield	nqdp disclosure	per policy edit (rare)	D6 Policy	per request
public config bitfield	nqdp disclosure	per policy edit; set-to-drop under attack	D6 Policy	per request
per-port mode (re-spond/relay/drop)	nqdp mode	per operator config	D3 Box mode	per request
local monotonic clock	hardware	continuous	D7 Assemble (responder epoch)	per request
response seq	—	—	—	— (removed; time replaces it)

The read side is always per-request and constant-time. The write side is always slow and asynchronous. No structure is both written and read on the per-request path, which is why there is no per-request contention and no per-request computation.

Appendix I — The Bundle Injected Per Prefix

The control plane assembles one bundle per prefix and injects it into `nqdp_prefix`. This is the full set of values a single lookup returns, their source, and how often the control plane refreshes each.

Bundle value	Source	Refreshed when
cost	carrier BGP RIB, verbatim	the prefix's BGP best-path cost changes
origin as seen	BGP AS PATH origin	best path changes
next hop as	BGP AS PATH first hop	best path changes
as path hash	plain hash over BGP AS PATH	AS PATH changes
as path len	BGP AS PATH length	AS PATH changes
rpki status	RPKI validator	validation state changes
reachable	presence of a usable best path	reachability changes
scrubbing active	mitigation/steering state	scrubbing engaged or disengaged
stability epoch	set to responder clock at each change	any best-path change
anycast modulus	this box's fixed ring position	box joins/leaves the ring
anycast count	ring membership size	membership changes

A miss on the `nqdp_prefix` lookup — a target the control plane has not populated — returns all sentinels. An unknown target is indistinguishable from a fully-squelched one, consistent with Rule 4.

Appendix J — Time-Series Behavior of Each Field

NQDP returns single-instant truth. Every rate, trend, and change is the requester's own derivation across stored rows. This appendix states what each field yields as a time series and the three-state read every field supports.

Every field has three distinct conditions in the series, none ambiguous:

Condition	On the wire	Meaning in the series
real value	a legitimate value	the fact at that instant
sentinel	all-ones	not available (withheld or unknown — indistinguishable)

Condition	On the wire	Meaning in the series
no row	no response arrived	the responder dropped, relayed with no answer, or was unreachable

Per-field series derivations:

Field	Single-row use	Across-row derivation
cost	pick lowest among responders	rising cost = drain or congestion; carrier steering
origin as seen	compare to expected origin	a change = possible hijack onset, timestamped
next hop as	who handles traffic now	a change = peering or path shift
as path hash	compare two vantages for asymmetry	a change = path changed; frequency of change = instability
as path len	compare route lengths	lengthening = detour, possible leak
rpki status	trust the route or not	valid→invalid transition = hijack or ROA problem
reachable	reachable now or not	yes→no transition = outage onset, timestamped
scrubbing active	mitigation in path now	off→on = attack detected and mitigated
stability epoch	time stable = epoch – stability epoch	repeated changes = flap frequency (the removed flap count)
anycast modulus	which instance now	changing across polls = instance flapping
anycast count	ring size now	shrinking = instances withdrawn, possible outage

The flap count removed from the packet (Section 6.3) reappears here as a derivation: the requester counts how often `stability_epoch` or `as_path_hash` changes per unit of its own wall-clock. The protocol carries the instant; the series carries the rate.

Appendix K — Attack-Surface Accounting

Every attack NQDP must resist, the mechanism that resists it, and where that mechanism lives.

Attack	NQDP resistance	Mechanism location
Reconnaissance (host existence)	reaching an instance $\neq$ confirming a hidden host; only chosen scalars disclosed	field design + disclosure config
OS fingerprinting	no response varies by host stack; one fixed record from a table	fixed response + injected bundle
Amplification	request 24 B, response 48 B, $\sim 1:1$, no multiplier	packet sizing
Resource exhaustion	answering is one table read at line rate	two-sided design (Appendix H)
Policy probing	withheld and unknown both = sentinel	sentinel rule
Flooding for advantage	flood absorbed as ordinary UDP by the forwarding plane	inherited forwarding-plane protection
Spoofed-source queries	SVT invalid $\rightarrow$ public audience; cannot gain trusted disclosure	inherited SVT (NET-1/NET-2)
Poisoning via client data	responder reads no client time, sequence, or identity	Rule 8
Replay / mismatched answers	request id echoed verbatim; requester matches on equality	identifier rule
Clock manipulation	responder supplies own monotonic time; never reads client time	Rule 8 + monotonic clock
Parser differential	fixed offsets, no options, one shape	Rules 1–2

The table shows NQDP’s own additions resist the diagnostic-specific attacks (reconnaissance, probing, poisoning, replay), while the volumetric attacks (flooding, spoofing, exhaustion) are resisted by inheriting the forwarding plane’s existing protections. NQDP does not re-implement DDoS defense; it rides on the one already built.

Appendix L — What NQDP Keeps, Drops, and Replaces from ICMP

Each ICMP function, and its disposition under IPv4-64 plus NQDP.

ICMP function	Disposition	Where it goes
Echo Request/Reply (ping)	Replaced	<code>anycast</code> <code>modulus/count</code> — reaching an instance is liveness
Time Exceeded (traceroute)	Dropped from forwarding; replaced at query level	<code>as path len / as</code> <code>path hash</code> give path facts without hop exposure
Destination Unreachable	Dropped	silent drop in forwarding; <code>reachable</code> field answers on authenticated request
Port Unreachable	Dropped	silent drop; no service probing surface
Redirect	Dropped	routing is control-plane only; no in-band redirect
Source Quench (deprecated)	Not carried	ECN in the IP header handles congestion signaling
Parameter Problem	Dropped	strict drop; malformed packets vanish silently
Fragmentation Needed (PMTUD)	Partially addressed	transit fragmentation (NET-1) reduces need; no ICMP black-hole dependence

The pattern: ICMP's error-signaling functions are dropped entirely because the forwarding plane reveals nothing, and ICMP's diagnostic functions are replaced by NQDP queries that answer the same operational question under carrier-controlled disclosure. The two roles ICMP fused — silent-drop error behavior and answer-anyone diagnostics — are cleanly separated: errors become silence, diagnostics become authenticated queries.

Appendix M — Connection to the Series Invariants

NQDP is the same design philosophy as the forwarding plane, applied one layer up. This appendix maps each series invariant to its NQDP expression.

Series invariant	Forwarding-plane form	NQDP form
Fixed size, no variable length	36-byte header, fixed TCP/UDP	24-byte request, 48-byte response
Worst case = best case	constant pipeline depth	one table read per query, always

Series invariant	Forwarding-plane form	NQDP form
Reveal nothing	strict silent drop	sentinels + disclosure config
Commensurate cost (parity)	expensive action gated behind proven spend	answering costs a table read, no amplification
Trust is bilateral and physical	SVT via meet-me-room secrets	SVT bit inherited; same trust root
Slow plane computes, fast plane serves	BGP injects routes, P4 forwards	control plane injects bundles, data plane answers
No interpretation of hostile input	fixed-offset reads, no parsing	responder reads no client claim about itself
Degrade without going dark	graceful drain via cost	slam-shut sheds public, keeps peers

The through-line: NQDP does not introduce a new security model, a new trust root, or a new performance model. It inherits all three from the forwarding plane and expresses them in the diagnostic layer. The reason the diagnostic service can be as safe and as fast as the forwarding plane is that it is built from the same parts.

Appendix N — Open Deployment Constants

Values a carrier sets at deployment. None affects the wire layout; all are local policy.

Constant	Set by	Consideration
UDP port	fixed at 56789	unregistered, uncommon; carriers may override by agreement
as path hash function	carrier, consistent across responders	plain, non-cryptographic; must match across boxes for cross-vantage comparison
responder epoch resolution	carrier	32-bit; choose seconds or finer against the rollover horizon
staleness horizon	requester	how old a cached responder epoch may be before re-polling
flap window	requester	wall-clock window over which stability epoch changes are counted

Constant	Set by	Consideration
per-box mode defaults	carrier	respond where table memory allows, relay elsewhere, drop at untrusted edges
disclosure bitfield defaults	carrier	which fields public sees; which SVT-only; drop flag for slam-shut
responder placement	carrier	respond-mode boxes need bundle-table memory alongside forwarding tables

[[HOWL-NET-1-2026](#)] IPv4-64: A 64-Bit In-Place Upgrade to IPv4 with Modern Security and Performance. Github: <https://github.com/ghowland/papers/tree/main/papers/NET/HOWL-NET-1-2026>

[[HOWL-NET-2-2026](#)] IPv4-64: Security and Performance Analysis. Github: <https://github.com/ghowland/papers/tree/main/papers/NET/HOWL-NET-2-2026>

[[HOWL-NET-3-2026](#)] IPv4-64 Reference Implementation on P4 Programmable Hardware. Github: <https://github.com/ghowland/papers/tree/main/papers/NET/HOWL-NET-3-2026>

[[HOWL-NET-4-2026](#)] IPv4-64 Corrected Header and Unified Routing Specification. Github: <https://github.com/ghowland/papers/tree/main/papers/NET/HOWL-NET-4-2026>